

Technical Report

Competition: The FREUID Challenge 2026 - IJCAI-ECAI

Team: kazihumayunrashid

Date: July 16, 2026

1. Introduction

The FREUID Challenge 2026 aims to detect next-generation identity document fraud across physical manipulations, GenAI-driven digital edits, and print-and-capture attacks. The task is binary fraud detection: 0 (bona-fide) versus 1 (fraudulent).

2. Method

Architecture: We use EfficientNet-B0 as the backbone feature extractor. The final classification layer is replaced with a custom head:

```
Dropout(0.5) → Linear(1280, 256) → ReLU → Dropout(0.3) → Linear(256, 1)
```

The output is passed through a Sigmoid function to produce a calibrated fraud probability in the range $[0, 1]$.

3. Data

- **Source:** FREUID 2026 official training dataset
- **Fraud Rate:** 42.32% (attack / fraudulent)
- **Preprocessing:**
 - Resize to 224x224
 - Normalize with ImageNet mean $[0.485, 0.456, 0.406]$ and standard deviation $[0.229, 0.224, 0.225]$

4. Training

- **Platform:** Kaggle Notebook (NVIDIA T4 GPU)
- **Loss Function:** Binary Cross-Entropy with Logits (BCEWithLogitsLoss)
- **Optimizer:** Adam
- **Model Format:** PyTorch .pth checkpoint

5. Inference

- **Batch Size:** 64
- **Test-Time Augmentation (TTA):** 5 augmented variants per image (original, horizontal flip, rotation ± 5 degrees, color jitter, combined) — predictions averaged.
- **Output Format:** CSV with columns `id, label` where `label` is fraud probability.

6. Results

Public Leaderboard Score: **0.28852**

Submission Rows: 142,818

7. Reproducibility

Docker Build:

```
docker build -t freuid-repro:local .
```

Docker Run (no network):

```
docker run --rm --network none \  
-v /path/to/test/images:/data:ro \  
-v "$(pwd)/out:/submissions" \  
freuid-repro:local
```

- **Random Seed:** 42
- **Hardware:** Training on Kaggle GPU (T4). Inference compatible with CPU and GPU.